

RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY
MODEL EXAMINATION (2022-23) ODD SEM
ENGINEERING THERMODYNAMICS

SECTION: A, B & C
DATE: 13.02.2023

PART – A

MAX.MARKS: 75
(10 x 2 = 20)

ANSWER ALL THE QUESTIONS

1. State the Zeroth law of thermodynamics.
2. Define pure substance.
3. What are the limitations of first law of thermodynamics?
4. What is perpetual motion machine of first kind?
5. State the Kelvin – Plank statement of second law thermodynamics.
6. Define COP?
7. Draw the PV & TS diagram for diesel cycle.
8. List the assumptions for air standard cycles
9. State a few properties of an ideal Refrigerant.
10. Compare vapour compression system and vapour absorption System.

PART – B

(5 x 11 = 55)

ANSWER ALL THE QUESTIONS

11. An engine cylinder has a piston area 0.12m and contains gas at a pressure o 1.5 MPa. The gas expands according to a process which is represented by a straight line on a pressure-volume diagram. The final pressure is 0.15 MPa. Calculate the work done by the gas on the piston if the stroke is 0.30m

(OR)

12. A gas of mass 1.5 Kg undergoes a quasi static expansion which follows a relationship $p = a + bV$, where a and b are constants. The initial and final pressures are 1000 KPa and 200 KPa respectively and the corresponding volumes are 0.20 m³ and 1.20 m³ The specific internal energy of the gas is given by the relation $U = 1.5 PV - 85 \text{ KJ/ Kg}$. Calculate the heat transfer of the gas during the expansion.
13. A stationary mass of gas is compressed without friction from and initial state pf 0.3 m³ and 0.105 MPa to a final state of 0.15 m³ and 0.105 MPa. The pressure remains constant during the process. There is a transfer of 37.6 kJ of heat from the gas during the process. How much does the internal energy of the gas changes?

(OR)

14. The internal energy of a certain substance is given by the following equations $U = 3.56 pv + 84$, where U is given in kJ/kg, p is in kPa, and v is in m³ /kg. A system composed of 3 kg

of this substance expands from an initial pressure of 500 kPa and a volume of 0.22 m^3 to a final pressure 100 kPa in a process in which pressure and volume are related by $p v^{1.2} = \text{constant}$. (a) if the expansion is quasi-static find Q , ΔU and W for the process. (b) in another process same system expands according to the same pressure-volume relationship as in part(a), and from the same initial state to the same final state as in part(a), but the heat transfer in this case is 30kJ. Find the work transfer for this process. (c) explain the difference in work transfer in parts (a) and (b).

15. A reversible heat engine operates between two reservoirs at temperatures 700°C & 50°C . The engine drives a reversible refrigerator which operates between reservoirs at temperatures of 50°C and -25°C . The heat transfer to the engine is 2500KJ and the net work output of the combined engine refrigerator plant is 400KJ. Determine the heat transfer to the refrigerant and the net heat transfer to the reservoir at 50°C .

(OR)

16. Two reversible heat engines A and B are arranged in series. A rejects heat directly to B. Engine A receives 200 kJ at a temperature of 421°C from the hot source while engine B is in communication with a cold sink at a temperature of 5°C . If the work output of A is twice that of B. Find the intermediate temperature between A and B, efficiency of engine, heat rejected to the sink.

17. Derive the efficiency of Otto cycle and mean effective pressure?

(OR)

18. In an air standard Diesel cycle, the compression ratio is 16, and at the beginning of isentropic compression, the temperature is 15°C and the pressure is 0.1MPa. Heat is added until the temperature at the end of the constant pressure process is 1480°C . calculate (a) the cut-off ratio, (b) the heat supplied per kg of air, (c) the cycle efficiency, and (d) the mean effective pressure

19. Explain the working principle of vapour compression refrigeration system with neat sketch

(OR)

20. Explain Claude system of liquefaction of gases with neat sketches.